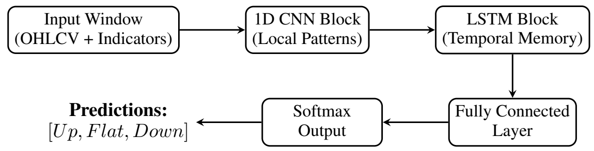

Figure 1: Architecture of the proposed hybrid CNN–LSTM model.

Cross-Regime Generalization in Cryptocurrency Direction Prediction: A Hybrid CNN–LSTM Stress Test

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https://github.com/Saberartoria77/APS-360-Project

1 Introduction

The mission for this project is simple but useful. In recent years, crypto has grown rapidly as a speculative asset class. Retail participants are drawn by outsized returns but exposed to extreme volatility and frequent losses: the market trades 24/7, has no earnings or dividends, and fraud is widespread, with participants at risk of catastrophic loss. This project reformulates the problem as a multi-horizon directional movement classification task (up, down, or flat) for major crypto assets (BTC and ETH). However, the core motivation of this project extends beyond standard prediction into evaluating model robustness. We therefore investigate the model's structural fragility under distribution shift—commonly known as market regime shift. By training the network in one market regime and evaluating it in a drastically different regime, we aim to expose and quantify the hidden performance decay that typically wrecks machine learning portfolios when deployed in production.

The experiment is designed for cross-regime generalization, motivated by evidence from **?**, who successfully applied an LSTM to directional prediction on the S&P 500 and found that it outperformed many other models in profitability. Crucially, however, their sub-period analysis revealed that the model's profitability deteriorated markedly across successive market regimes (shifting from 1993–2000 to 2001–2009 and finally to 2010–2015). Rather than treating this degradation as a tragic footnote, our project elevates it to the primary object of study. Using high-frequency data extracted via the Binance Public API, we will systematically partition our dataset into distinct, mathematically defined regimes to chart exactly how, and why, a competent deep learning model's predictive edge dissolves when the statistical properties of the world change.

Cryptocurrency price signals are inherently non-linear and non-stationary, meaning their statistical properties change over time. Traditional linear models, such as ARIMA or standalone logistic regression, assume overly simplistic relationships and fail to capture these complex dynamics. To address this, our approach utilizes a hybrid deep learning architecture. A 1-dimensional convolutional neural network (1D CNN) acts as a feature extractor, sliding over recent price windows to identify local, short-term temporal patterns. Because a CNN cannot retain long-term sequential context, the extracted features are fed into a long short-term memory (LSTM) network, which captures the temporal dependencies of ordered time-series data. Together, they represent sequential financial data significantly better than either architecture in isolation [**?**]. Crucially, it is exactly this high expressivity that makes the CNN–LSTM the perfect subject for our investigation: because the model is flexible enough to tightly fit a specific market regime, it is equally prone to catastrophically overfitting it. This structural fragility is precisely what we intend to stress-test.

Preprint.

2 Background & Related Work

The foundation for directional financial prediction using deep learning was established by **?**, who demonstrated that LSTMs consistently outperform memory-free models on equity markets, though their sub-period analysis revealed profitability degradation across distinct market eras. **?** further validated the use of LSTMs specifically for predicting asset price movements using standard technical indicators. Subsequent research successfully adapted these methods to the cryptocurrency domain; **?** utilized LSTMs on trade data to achieve predictive accuracy high enough to be monetizable, while **?** validated the standard use of logistic regression and random forests as robust financial baselines. Expanding on sequence modeling, **?** proposed a hybrid CNN–LSTM framework for Bitcoin, proving that combining a CNN's local feature extraction with an LSTM's temporal memory significantly outperforms either architecture in isolation. Our project builds on this lineage, using the hybrid structure of **?** to investigate the structural decay hinted at by **?**.

3 Data Processing

We will source high-frequency OHLCV (open, high, low, close, volume) data for BTC and ETH using the public Binance API (`/api/v3/klines`). To enrich the feature space, we will compute standard technical indicators such as RSI, MACD, and Bollinger Bands. To strictly prevent look-ahead bias, all features will be normalized using z-scores calculated exclusively from the training-set distribution. The continuous data will be sequenced into sliding windows of approximately 96 hours. Most importantly, to support our core hypothesis regarding market regime shift, the dataset will be partitioned chronologically into distinct, volatility-defined market phases (e.g., low-volatility consolidation versus high-volatility drawdowns) rather than randomized splits, ensuring a rigorous test of cross-regime generalization.

4 Architecture

The proposed model processes a sliding window of historical cryptocurrency data, forming an input tensor of size $T \times F$, where T represents the time-steps (e.g., 96 hours) and F represents the features. This sequence is first fed into a 1D CNN block with ReLU activations to extract rich local temporal patterns from short-term price movements. Crucially, the CNN preserves the temporal sequence without condensing it. The LSTM then reads that enriched sequence in order, carrying memory forward, and condenses all timesteps into a single summary vector that retains temporal dependencies. Finally, this summary vector is routed through a fully connected output layer applying a softmax activation to yield probabilities for the three directional classes: up, flat, and down. The architecture will be implemented in PyTorch, using cross-entropy loss and the Adam optimizer, with hidden dimensions tuned empirically on a validation set.

5 Baseline Models

To contextualize the deep learning model's performance, we will implement two non-sequential baselines. The first is a naive momentum heuristic that predicts the direction of the most recent time-step. The second is a logistic regression model, implemented via scikit-learn, trained on the same engineered feature set. These memory-free baselines quantify the specific predictive value added by the CNN–LSTM's temporal modeling.

6 Ethical Considerations

Financial machine learning models carry substantial ethical risks, particularly for retail participants who face extreme volatility and potential catastrophic loss. Models that show high backtest accuracy frequently collapse in production due to distribution shifts, creating a false sense of security that misleads investors and amplifies market harm. By explicitly designing this project to measure how a model's predictive edge degrades across market regimes, we address the imperative of honest evaluation and highlight the dangers of deploying brittle, overfitted systems in real financial environments without cross-regime stress-testing.

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